

Experiment No. 11

Title

Depth First Search (DFS) Using Artificial Intelligence

Aim

To develop a Python program to implement the Depth First Search (DFS) algorithm for graph traversal.

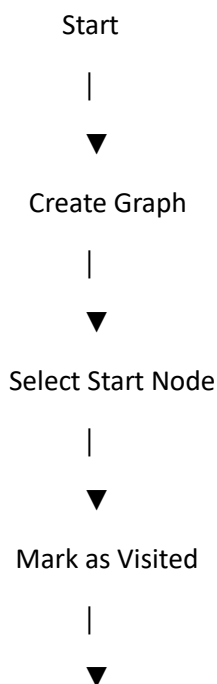
Objective

The objective of this experiment is to understand the Depth First Search (DFS) algorithm and implement it to traverse a graph by visiting nodes in a depth-wise manner.

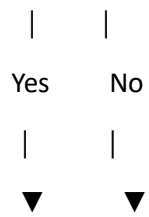
Algorithm

- Step 1:** Create the graph using an adjacency list.
 - Step 2:** Select the starting node.
 - Step 3:** Mark the starting node as visited.
 - Step 4:** Visit each unvisited adjacent node recursively.
 - Step 5:** Continue until all nodes are visited.
 - Step 6:** Display the traversal order.
 - Step 7:** Stop.
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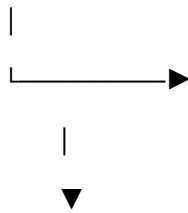
Flowchart



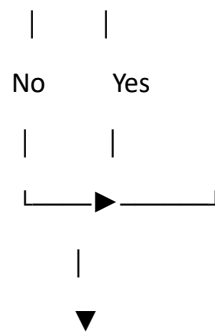
Unvisited Neighbour?



Visit Neighbour Return



All Nodes Visited?



Display Traversal



Python Program

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}
```

```
visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

    for neighbour in graph[node]:
        dfs(neighbour)

print("DFS Traversal:")
dfs('A')
```

Sample Output

DFS Traversal:
A B D E F C

Result

The graph was successfully traversed using the Depth First Search (DFS) algorithm.

Conclusion

Thus, the Depth First Search (DFS) algorithm successfully traversed the graph by exploring each branch completely before backtracking. It is an efficient search technique for graph traversal and path exploration.